

An Empirical Decomposition of Information vs Sentiment Shifts in Advertising^{*}

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Abstract

Advertising influences market expectations through three distinct pathways: raising awareness, shaping sentiment, and providing information. This paper empirically disentangles these effects using movie trailers as an empirical setting, leveraging a dataset of nearly 600 wide-release U.S. films (2017-2021). We analyze how trailer releases affect pre-theatrical demand expectations, as reflected in Hollywood Stock Exchange (HSX) prices, and measure the three channels of consumer engagement through Twitter discussions and Wikipedia page views. Our findings show that trailers significantly boost HSX prices, particularly for non-sequels with lower prior awareness. While heightened engagement correlates with stronger market expectations, additional informational content can sometimes dampen enthusiasm. Moreover, trailer releases reduces uncertainty, as evidenced by a decline in HSX price volatility for non-sequels. By disentangling the three pathways of advertising influence, this study contributes to the literature on advertising effectiveness and offers actionable insights for marketers optimizing promotional strategies in a digital landscape.

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1 Introduction

Advertising plays a critical role in shaping consumer perceptions and market behavior. The literature identifies three distinct mechanisms through which advertising influences consumer behavior and market expectations. First, it raises awareness by increasing a product’s visibility, a crucial function in markets with frequent new product launches. Second, it shapes sentiment, evoking emotional responses such as humor, nostalgia, excitement, or fear-particularly in industries where differentiation hinges on perception rather than tangible attributes. Finally, advertising conveys specific information about a product’s features, quality, or benefits, relying on explicit knowledge transfer rather than emotional appeal.

A key contribution of this paper is the empirical decomposition of advertising’s impact into three distinct pathways: raising awareness, shaping sentiment, and conveying information. Disentangling these effects is challenging because they often work in tandem to drive demand, particularly in industries with minimal price competition, abundant product offerings, and opaque advertising expenditures. Many sectors face this complexity. In pharmaceutical industry, direct-to-consumer advertising raises awareness of new treatments while also shaping brand perceptions. Luxury fashion brands cultivate exclusivity and desirability through advertising, influencing sentiment and awareness without explicit price-based competition. The video game industry leverages influencer partnerships and promotional events to generate excitement, blurring the line between informational and emotional effects. Understanding how these mechanisms operate is crucial for businesses, investors, and policymakers who seek to optimize marketing strategies and predict consumer behavior more effectively.

This paper examines the movie industry, focusing on trailer releases as a natural setting to study advertising effects. Movie trailers serve as distinct and measurable promotional events, making it easier to isolate their impact on consumer behavior.¹ Unlike many other forms of advertising, the timing of trailer releases are largely a surprise to con-

¹Trailer releases are pivotal moments in a film’s promotional campaign. Studies show that movie trailers significantly impact a movie’s success. Boksem and Smidts (2015) provide evidence using both stated preference and neural measures. Karray and Debernitz (2017) show that the appeal of the movie plot, the number of scene cuts, and the inclusion of violent, sexual, or humorous scenes in a trailer all influence a movie’s abnormal returns.

sumers, minimizing anticipation bias and allowing for precise measurement of real-time shifts in consumer responses. This unique characteristic enables researchers to assess how trailers influence demand expectations, consumer awareness, and sentiment toward a film. Trailers not only generate excitement and shape sentiment but also provide key informational content—revealing a film’s genre, cast, storyline, and production quality. A second advantage of our focus is in the product space: we can separately analyze incremental products with are slight variants of pre-existing ones (sequels or prequels) and more novel products about which there is limited consumer awareness and informaton (non-sequels). By analyzing consumer reactions before and after trailer releases, this study offers insights into how these promotional materials shape market behavior and influence the overall success of a film.

This paper investigates how advertising (movie trailers) empirically influence market expectations through three pathways. Specifically, we analyze nearly 600 wide-release U.S. films from 2017 to 2021, leveraging data from Twitter’s movie-related archive, Wikipedia pageview histories, and Hollywood Stock Exchange (HSX) daily prices. We use the HSX, a long-running prediction market where participants trade shares linked to the revenue of upcoming films, to measure real-time expectations of theatrical box office demand. By tracking social media activity, sentiment in Twitter discussions, Wikipedia page views, and informational content updates, we assess how trailer releases shape consumer awareness, sentiment, and investment behavior, as reflected in pre-theatrical HSX price movements.

The HSX provides a real-time measure of box-office expectation before theatrical releases. The HSX serves as a dynamic aggregator of market beliefs, rapidly incorporating new information through trading activity. Prior research has shown its accuracy in forecasting box office revenues, making it a valuable tool for measuring demand expectations. The availability of the HSX is one reason we focus on the movie industry, as few other contexts offer such a product-specific and continuously updated measure of expected sales.

We use Twitter and Wikipedia activity as key indicators of consumer engagement and awareness in response to movie trailer releases. Twitter data serve two distinct purposes in our analysis: (1) the volume of original tweets about a movie is used as a proxy for public awareness, and (2) textual analysis of tweet content provides a measure of consumer

sentiment toward the film and information about the film. Using the Twitter API, we collect all original tweets related to each wide-release film, assembling a dataset of 10 million tweets, averaging 17,000 tweets per movie, with details on text, retweets, and likes to capture both volume and sentiment. Similarly, Wikipedia page views serve as an indicator of consumer interest and information-seeking behavior. Extracted via the Wikipedia API, these data reveal spikes in consumer interest following trailer releases. Additionally, changes in Wikipedia content before and after a trailer release provide a valuable proxy for the informational content conveyed by the trailers.

Our findings reveal that advertising releases generate significant positive shifts in market expectations, as reflected in changes in HSX prices. Trailer releases trigger immediate and substantial price increases, with effects more pronounced for novel products (non-sequels) than for familiar ones (sequels). Non-sequels see HSX prices increase by over 10 percent, while sequels experience a more modest increase of around 4 percent. This suggests that advertising is particularly crucial for products with lower prior awareness.

We further examine how advertising influences market expectations through awareness, sentiment, and informational effects. Higher tweet volume and Wikipedia page views strongly correlate with HSX price increases. A one standard deviation increase in tweet volume following a trailer release corresponds to a 3 percent rise in HSX price, while a similar increase in Wikipedia page views leads to a 6 percent increase. However, sentiment and informational content exhibit more nuanced effects. More positive tweet sentiment is associated with a modest increase in HSX price, whereas an increase in detailed informational content – in terms of Wikipedia page contents – is linked to a small but statistically significant decline in HSX price. This suggests that while positive sentiments boost movie expectations, more detailed information can sometimes temper enthusiasm.

We also find that advertising helps reduce market uncertainty, particularly for novel products. Following trailer releases, HSX price volatility declines significantly for non-sequels, suggesting that trailers help resolve uncertainty about the commercial prospect of relatively unknown films. Consistent with our interpretation, after a trailer release price volatility for non-sequels converges with sequels (the latter has more information and lower volatility prior to the trailer). The reduction in volatility for non-sequels reflects

more than just informational updates, it also captures the market’s response to a clearer commitment by the studio. Releasing a trailer signals that the studio intends to proceed with a theatrical release, as opposed to pivoting to non-theatrical distribution channels such as streaming or pay-per-view, which remain common alternatives. These distribution decisions are often finalized only a few months before a film’s planned release, especially for less-established titles. Therefore, trailers serve not only to raise awareness, shape sentiment, and convey content but also to act as credible signals of commitment to a theatrical run. This commitment reduces ambiguity for both consumers and investors, particularly for products with limited prior information and higher perceived risk.

This research contributes to the broader literature on advertising and promotion by providing an empirical framework to disentangle the intertwined effects of awareness and persuasion. By integrating insights from marketing and digital media research, this study enhances the understanding of how advertising signals shape market expectations and consumer behavior across industries. The findings highlight advertising’s dual role— not only in engaging audiences but also in shaping confidence in a product’s commercial success.

This paper also provides actionable insights to managers and industry practitioners. While promotional activities influence consumer knowledge and perceptions, these effects do not always translate directly into shifts in demand expectations. Our results reveal the complex ways in which advertising can amplify – or fail to amplify – the responsiveness of demand to changes in awareness and sentiment. For practitioners, this paper underscores the importance to design promotional strategies that not only generate buzz but also effectively convert consumer interest into demand. By leveraging real-time data and exogenous shocks, we provide evidence-based insights into how advertising drives measurable outcomes in consumer markets, offering guidance for marketers, product managers, and policymakers navigating in an increasingly digital landscape.

The remainder of the paper is as follows. Section 2 reviews the related literature. Section 3 describes the data, while Section 4 provides an informal overview of movie trailer impacts. Section 5 has the main analysis and empirical results. Section 6 concludes, and the Appendices contain robustness checks.

2 Literature

Since Chamberlin (1933), research has identified the primary effects of advertising on demand: informational and persuasive. The persuasive role, highlighted by Robinson (1933) and others (Braithwaite, 1928; Kaldor, 1950; Bain, 1956), suggests that advertising shapes consumer preferences, potentially impacting market competition. The informational role, advanced by Ozga (1960) and Stigler (1961), suggests that advertising conveys essential product details, reducing price dispersion and promoting competitive behavior (Telser, 1964; Nelson, 1970, 1974). Our study builds on this literature by empirically demonstrating that movie trailers perform all these functions: raising awareness, shaping sentiment, and providing information about a film’s genre, cast, and storyline. Crucially, we show that these effects do not uniformly translate into higher demand expectations. In some cases, greater exposure or more detailed information dampens market enthusiasm. By analyzing how these factors influence shifts in consumer demand, our research highlights the nuanced and multidimensional impact of advertising, both within the movie industry and in broader market contexts.

Our paper also contributes to the literature of the impact of word-of-mouth in shaping consumer demand. Even before the emergence of the internet and social media, word-of-mouth played a crucial role in determining box office performance (De Vany and Walls, 1996; Moul, 2007). The digital age has amplified both the scale and complexity of word-of-mouth dynamics. For instance, while volumes of tweets and online reviews significantly influence sales prior to the launch of a product (Marchand, Hennig-Thurau, and Wiertz, 2017), the valence—rather than volume—tend to matter more after launch (Chintagunta, Gopinath, and Venkataraman, 2010; Marchand, Hennig-Thurau, and Wiertz, 2017). Among online reviews, consumers often view negative word-of-mouths as more reliable, perceiving it as less biased by promotional efforts (Chevalier and Mayzlin, 2006; Dellarocas, 2006). Our study extends this literature by quantifying the impact of trailers on word-of-mouth through Twitter and Wikipedia, capturing how trailer-driven discussions shape consumer engagement and market expectations in real time.

3 Data

This section describes the construction of our dataset, which integrates information from multiple publicly available sources to capture movie characteristics, public awareness, sentiment, informational content, and market expectations. We also address issues related to our observation window and sample period.

3.1 Movie Data

Our movie dataset includes all 580 movies that received a wide release in the U.S. market between 2017 and 2021. Most movie-level information was collected from www.boxofficemojo.com, a subsidiary of IMDb (owned by Amazon). This dataset contains information on each movie’s genre, theatrical release date, and whether a movie is a sequel/prequel within an existing franchise.

3.2 Twitter Data

Twitter plays a crucial role in shaping movie-related discussions, serving as a key platform for sharing news, reviews, and audience reactions. We use Twitter to measure consumer awareness, sentiment and information about each movie in our data. Studies have shown that social media buzz, particularly on Twitter, significantly influences consumer interest and box office performance by amplifying word-of-mouth effects. Asur and Huberman (2010) find that the volume and sentiment of tweets before a movie’s release strongly predict its box office revenue, often outperforming traditional forecasting models. Similarly, Rui, Liu, and Whinston (2013) show that social media discussions, including Twitter activity, can affect movie sales by influencing public perception and consumer decision-making. Additionally, Twitter’s trending algorithms boost the visibility of highly discussed movies, reinforcing its importance as a strategic marketing channel for studios seeking to drive audience engagement.

In January 2021, Twitter granted academic researchers full access to its tweet archive. We use the Twitter API to extract all original tweets (excluding retweets) containing hashtags associated with the movies in our sample, posted within 120 days prior to each

movie’s release (we discuss the observation window in Section 3.6). For each tweet, we collect the content, timestamp, number of likes and retweets, and user information such as the username and Twitter ID. Full-archive access for academic researchers was discontinued following Twitter’s ownership change in late 2022.

Our dataset comprises approximately 10 million tweets across 580 movies, averaging 17,000 tweets per movie. Except for a few movies (less than 3% of the sample) with zero tweets, the distribution of the number of tweets for each movie is log-normal. As we discuss later, tweet volume will help measure consumer awareness of a movie.

We use textual analysis to construct a sentiment score for each tweet. Specifically, we use the VADER sentiment analysis tool (Hutto and Gilbert, 2014) from the Natural Language Toolkit (NLTK) in Python (Bird, Klein, and Loper, 2009) to implement sentiment analysis.² VADER draws on established lexicons such as LIWC, ANEW, and GI, and is designed to handle social media-specific language, including emoticons and cultural acronyms (e.g., LOL), making it particularly well-suited for analyzing tweet content.

VADER constructs sentiment scores by first assigning each word in the text a valence score drawn from a pre-built dictionary. These individual scores are then adjusted based on linguistic rules and aggregated. The overall sentiment of the text – known as the compound scores – is computed by normalizing the sum of the adjusted valence scores to a range between -1 (most negative) and +1 (most positive). The daily sentiment measure for a movie is calculated by averaging the compound scores of all tweets that include the movie’s hashtag on a given day. In our sample, approximately 6% of movie-days have only neutral tweets, resulting in a compound score of zero. The mean daily sentiment score is 0.15, indicating a slightly positive tone.

To quantify the informational content of tweets, we examined how movie-related discussions evolved on a daily basis. Specifically, we calculated the proportion of tweet text that mentioned key figures such as actors, directors, and writers. For example, in the tweet “The #BossBaby Trailer: Alec Baldwin is a Baby on a Mission,” the actor’s name comprises 11 of the 48 characters, resulting in an information proportion of 0.23. We

²As Hutto and Gilbert (2014) shows, VADER incorporates syntax and punctuation rules and is validated with human coding, achieving sentence-level prediction accuracy between 55% and 96% accurate—comparable to more complex models such as the Stanford Sentiment Treebank.

computed this measure daily across all movies to track shifts in shared informational content. On average, about 15 percent of movie-day observations contained no information at all. Among the remaining observations, the distribution of information per tweet is approximately log-normal, with a median value of 0.05.

3.3 Wikipedia Data

We use a movie’s Wikipedia page views and page length to measure awareness and information, respectively.³ Wikipedia is a valuable pre-release information source, offering audiences details on casting, production, plot summaries, and promotional updates. Unlike traditional marketing channels, Wikipedia content is collaboratively edited and updated in real time, often incorporating new information as it emerges. Mestyán, Yasseri, and Kertész (2013) demonstrates that Wikipedia activity, including page edits and views, can serve as an early predictor of box office performance. Their study finds that higher engagement on a movie’s Wikipedia page before release is associated with stronger financial performance, highlighting Wikipedia’s role in shaping audience expectations and signaling consumer interest.

According to an industry insider who recommended this measure (and who played a prominent role in refining audience tracking surveys), movie studios use Wikipedia page views to gauge consumer awareness. Nearly all but the very smallest movies have dedicated Wikipedia pages, with separate entries often created for different versions of the same title or for adaptations from other media, such as novels.⁴

These pages are typically created around when the title is green-lit for production (or earlier for sequels). There are typically few page views until a trailer is released, unless there is other reasons for titles awareness such as the casting of a major star.

Similar to Tweet counts, the total number of Wikipedia page views per movie follows a log-normal distribution. The median number of page views is approximately 440,000.

³We use the English language Wikipedia, <https://en.wikipedia.org>. For page views we exclude spider and automated traffic.

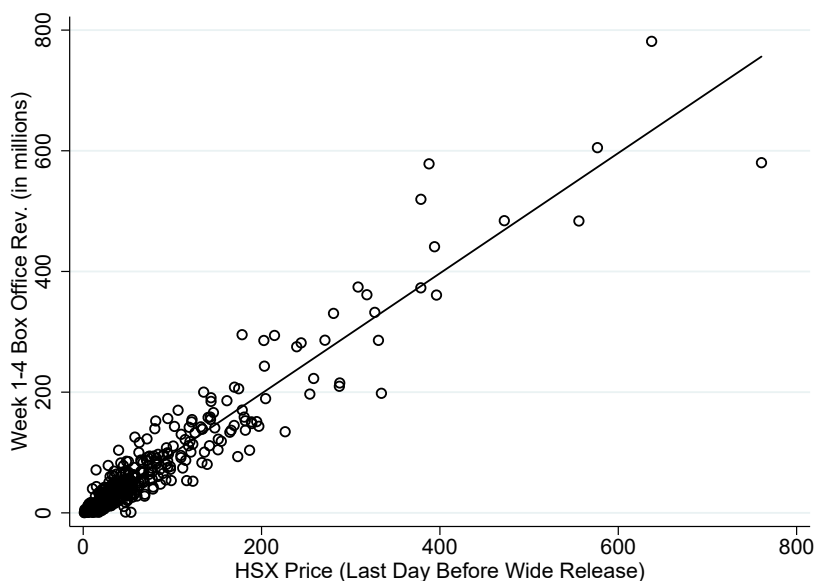
⁴For each movie we use the Wikipedia API to find potential matches within the “Category:Film” or similar categories. Each Wikipedia page match was manually checked. Three percent of the titles in the full sample do not have dedicated Wikipedia pages, and none of these titles have HSX tracking stocks and so were not included in the main analysis.

To capture the volume of publicly available information, we measure each movie’s Wikipedia page length in words and track its daily changes. For example, prior to the release of *The Boss Baby* trailer, the film’s Wikipedia page included only basic production details. Following the trailer’s debut, edits added descriptors such as “family buddy comedy,” concrete information about the movie’s tone and genre. On average, a movie’s Wikipedia page contains approximately 4,000 words across all movie-day observations.

3.4 Hollywood Stock Exchange (HSX)

The Hollywood Stock Exchange (HSX) is a movie-focused prediction market where participants use simulated money to trade “stocks” of upcoming films. It has a strong track record of forecasting accuracy and quickly incorporating new information. We use the HSX to measure real-time expectations of movie demand: theatrical box office.⁵

Figure 1: Correlation between HSX price and Box Office Revenue



We collected daily HSX data for each movie in our sample, covering end-of-day stock prices before theatrical release and continuing until the contract expires – typically one to three months after release. Figure 1 shows that a movie’s HSX price on the day before its

⁵Spann and Skiera (2003) show that HSX predictions compare favorably to other expert judgments in the movie industry.

theatrical release is highly predictive of its four-week box office revenue, with an almost one-to-one correlation (based on the movie sample described in Section 3.1).⁶

HSX is not only a strong predictor of box office performance but also rapidly incorporates new information. As news emerges, HSX prices adjust in ways that reflect the nature and tone of that information. Elberse (2007), using daily HSX data similar to ours, finds that prices respond significantly to casting announcements, supporting the idea that star involvement positively influences revenue expectations. Strumpf (2014) shows that HSX prices also react to trailer releases—well-received trailers lead to significant price increases (e.g., an 11-point rise for top-rated trailers), while poorly received ones cause notable declines (e.g., a 6-point drop). In short, movements in HSX prices provide a useful signal of whether a film has received favorable or unfavorable news.

3.5 Movie Trailers

We collect trailer release information by reviewing the discussion boards for each movie on the HSX website, focusing on posts announcing trailer debuts. These posts typically include trailer release dates, which we record for analysis. To ensure accuracy, we cross-check the dates using multiple trailer listing websites and a curated list of official trailers on YouTube. Our dataset includes all trailers released within 120 days prior to theatrical release (we discuss the observation window in Section 3.6). Approximately 80% of the movies in our sample had at least one trailer released within four months of their theatrical debut, and about 20% had multiple trailers during this period.

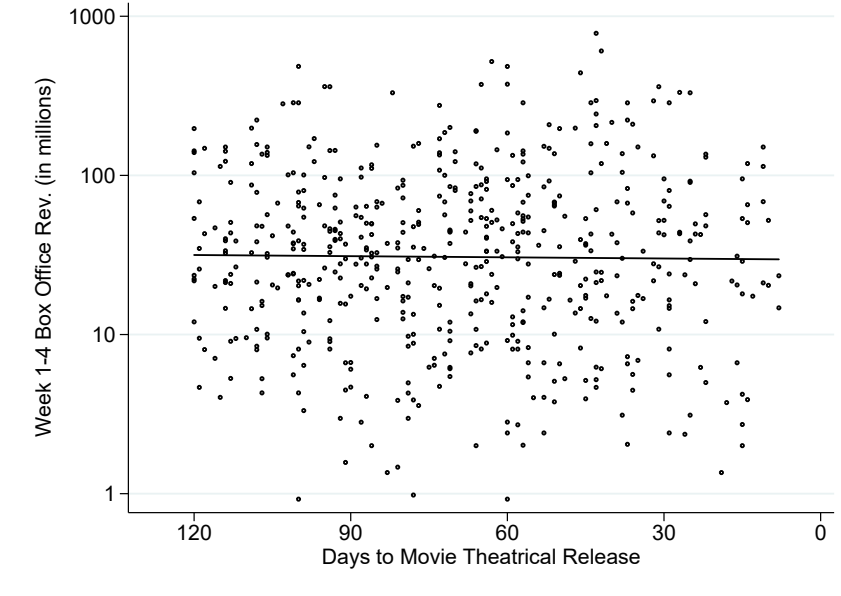
One potential concern in our analysis is the endogeneity of trailer release timing. Studios may strategically schedule trailer releases to maximize box office impact, introducing selection bias and undermining the assumption that trailer releases are exogenous.

To address this issue, we examine the relationship between trailer timing and subsequent box office performance. Figure 2 plots this relationship, with the x-axis representing the number of days between the release of the first trailer and the movie’s theatrical debut, and the y-axis showing the natural logarithm of box office revenue over the first

⁶Pennock et al. (2000) report similar findings using a sample of 109 movies between March and September 2000. In our data, an OLS regression of box office revenue on pre-release HSX price yields a coefficient of 1.005 (compared to their 1.004), with a standard deviation of 0.011.

four weeks. The figure shows no clear correlation between trailer timing and box office outcomes, suggesting that trailer release dates do not systematically reflect expectations of a movie’s success.

Figure 2: Correlation Between Trailer Release and Box Office Revenue



3.6 Observation Window and Sample Period

Our main analysis focuses on trailers released within 120 days of the theatrical release. The primary reason for focusing on the 120-day window before a movie’s release is due to data availability constraints. Specifically, our Twitter data is limited to tweets posted within 120 days prior to each film’s release date. Due to changes to Twitter’s API access policy, we can no longer retrieve historical tweet data beyond this window.

Appendix A has a variety of robustness checks that this does not bias our results. The main estimates are largely unchanged when we consider specifications which include earlier trailer releases. The estimates also remain similar when only trailers close the theater date are used, showing the results are not sensitive to the time window. Finally, we show there are no significant observable differences for movies releasing a trailer prior to our observation window.

A second timing issue is that our sample period includes 2020 and 2021 when many

theaters were shut due to COVID. This led to reduced box office and rescheduling of theatrical releases. To account for COVID-related uncertainty, we conduct a robustness check restricting the sample to pre-March 2020 releases. Appendix B shows the main estimates are largely the same for this sub-period.

4 Empirical Summary of Trailer Release Effects

This section summarizes and provides descriptive statistics of the effect of trailer releases on digital engagement, informational content, consumer sentiments, and market expectations. It also highlights how these effects differ for non-sequels and sequels.

4.1 Trailer Releases and Changes in Key Variables

We assess the impact of movie trailer releases on online engagement, sentiment change, information content, and market expectations by analyzing changes in key variables: daily tweet volume, Wikipedia page views, tweet sentiment, information content per tweet, Wikipedia Length, and HSX prices. To do so, we track daily changes and compare these variables between trailer release days and non-release days.

Table 1 reports summary statistics for the changes in each variable. All values reflect log changes, except for Tweet Sentiment and Information per Tweet. All log changes, except for HSX prices, are zero-inflated.⁷ Tweet Sentiment is measured as the change in the actual sentiment score. Information per Tweet is measured as the change in the proportion of tweet information content. Columns 1 and 2 present the mean changes on non-trailer and trailer release days, respectively. Column 3 reports the difference in means between the two types of days.

Notably, HSX prices show little movement on non-trailer days but increase by an average of more than 12 percent on trailer release days. This suggests that trailers prompt a meaningful market reassessment of a movie’s prospects in response to new promotional content.

⁷We apply the transformation $\log(x + 1)$ to retain zero-valued observations, which would otherwise be excluded under a standard logarithmic transformation. This approach preserves the full distribution while allowing for log-scale analysis.

Table 1: Trailer vs. Non-Trailer Day Summary Statistics

Changes	Non-Trailer Days	Trailer Days	Difference
HSX price	0.000 (0.001)	0.123 (0.009)	0.123*** (0.002)
Tweets Counts	0.031 (0.004)	2.666 (0.089)	2.635*** (0.040)
Wiki Page Views	0.019 (0.001)	1.417 (0.050)	1.398*** (0.014)
Tweet Sentiment	0.002 (0.001)	-0.012 (0.010)	-0.014 (0.008)
Information per Tweet	0.000 (0.000)	0.047 (0.005)	0.047*** (0.003)
Wiki Page Length	0.011 (0.001)	0.112 (0.034)	0.100*** (0.009)
Observations	67,417	577	67,994

All values reflect log changes, except for Tweet Sentiment and Information per Tweet.

Tweet Sentiment is measured as the change in the actual sentiment score.

Information per Tweet is measured as the change in the proportion of tweet information content.

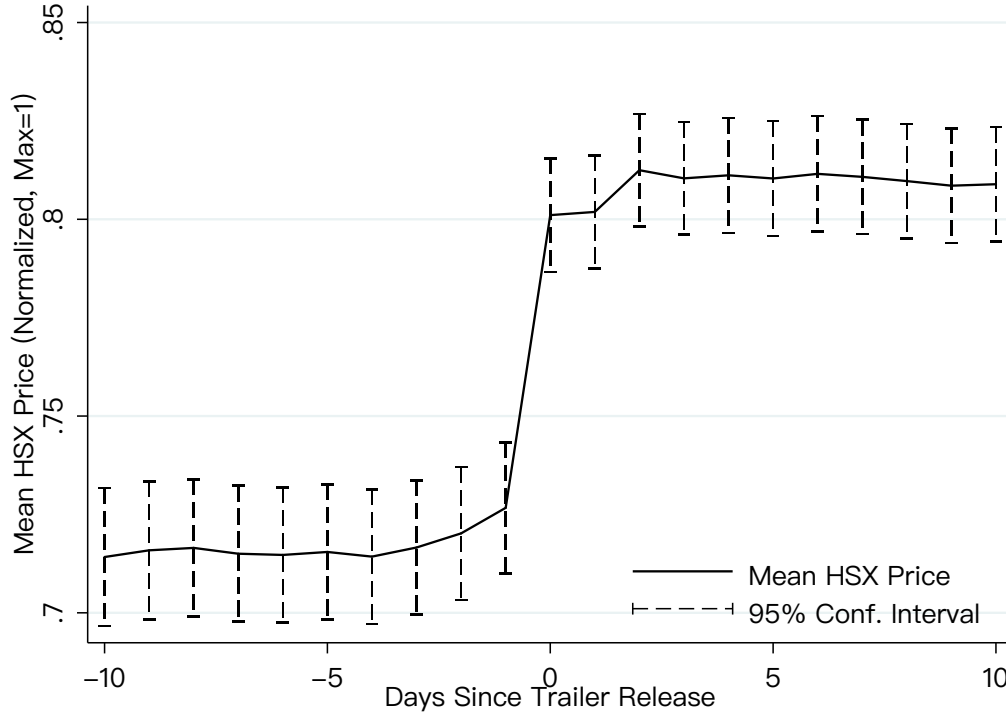
Standard errors in parentheses for all columns. * $p < .1$, ** $p < .05$, *** $p < .01$

Public engagement also spikes sharply on trailer release days. Twitter volume rises dramatically compared to non-trailer days, while Wikipedia page views also show a significant increase. These changes highlight the broad digital impact of trailers in amplifying audience attention and stimulating information-seeking behavior.

The information content per tweet increases significantly on trailer release days, by about 5 percentage points, indicating more specific mentions of movie elements such as cast or directors. Wikipedia page length grows slowly on non-trailer days, increasing by about 1% on average. However, on the days of trailer release, the length growth exceeds 10%, reflecting the addition of new details and edits in response to the trailer release.

Changes in tweet sentiment remain statistically insignificant on both non-trailer and trailer days, although the variance in sentiment is larger on trailer days. This likely reflects the range of reactions trailers can evoke, from excitement to disappointment, which cancel out in aggregate and result in stable sentiment measures.

Figure 3: Changes in HSX Price by Days since Trailer Release



While Table 1 highlights average differences between trailer and non-trailer days, it does not capture how these variables evolve around the time of a trailer release. To examine the temporal dynamics of market expectations, Figure 3 presents the average normalized HSX price over a 20-day window—spanning the 10 days before and after a trailer drop (Figure 4 has comparable time series for other variables). To ensure comparability across films, we normalize each metric by dividing daily values by the movie-specific maximum, bounding them between 0 and 1. This standardization enables meaningful comparisons across movies with different levels of popularity and market visibility.

Figure 3 shows that HSX prices remain steady in the days leading up to a trailer release, averaging around 72% of their peak value. On the day of the release, prices jump to over 80%, and this gain persists over the following 10 days. The increase of approximately 11% ($= 80\%/72\% - 1$) aligns closely with the average difference reported in Table 1.

This consistent price jump around trailer releases raises a potential puzzle. One might expect that rational investors would exploit such a predictable trading opportunity. For

example, they could hold a diversified portfolio of movie stocks and sell each one immediately after a trailer release to capture systematic gains. In theory, if enough investors pursued such arbitrage strategies, trailer-day price increases would eventually be largely eliminated.

However, this does not occur because the trade strategy is not riskless and can result in losses or no gains. A substantial share of films listed on HSX never make it to theaters, and so their HSX stocks expire worthless resulting in losses for their owners. This can even happen with completed or nearly finished movies: recent examples of major studio projects that were ultimately pulled from theatrical release include Martin Scorsese’s *The Irishman* (2019), *Batgirl* (2022), *Coyote vs. Acme* (2023), *War of the Worlds* (2024), and *Horizon: An American Saga – Chapter 2* (2024). In this sense the stocks are like traditional financial options, with positive values only if the movie has a theatrical release. Trailers are commitment devices signaling such a favorable expiry, but the key point is that such an outcome is uncertain at the time when the trade strategy would be implemented.⁸

Figure 4 tracks other key variables in a similar way to examine how they evolve around trailer releases. As shown in Figure 4a, Twitter activity spikes sharply on the day of a trailer release, while Wikipedia page views rise more gradually, peaking one day later. These patterns reflect the substantial differences reported in Table 1, confirming that trailers generate an immediate surge in public awareness. Unlike HSX prices, which remain elevated after the trailer release, tweet volume and Wikipedia page views decline following their initial spike. Tweet activity returns to pre-trailer levels within five days, while Wikipedia page views also decrease but stay higher than their pre-release baseline.

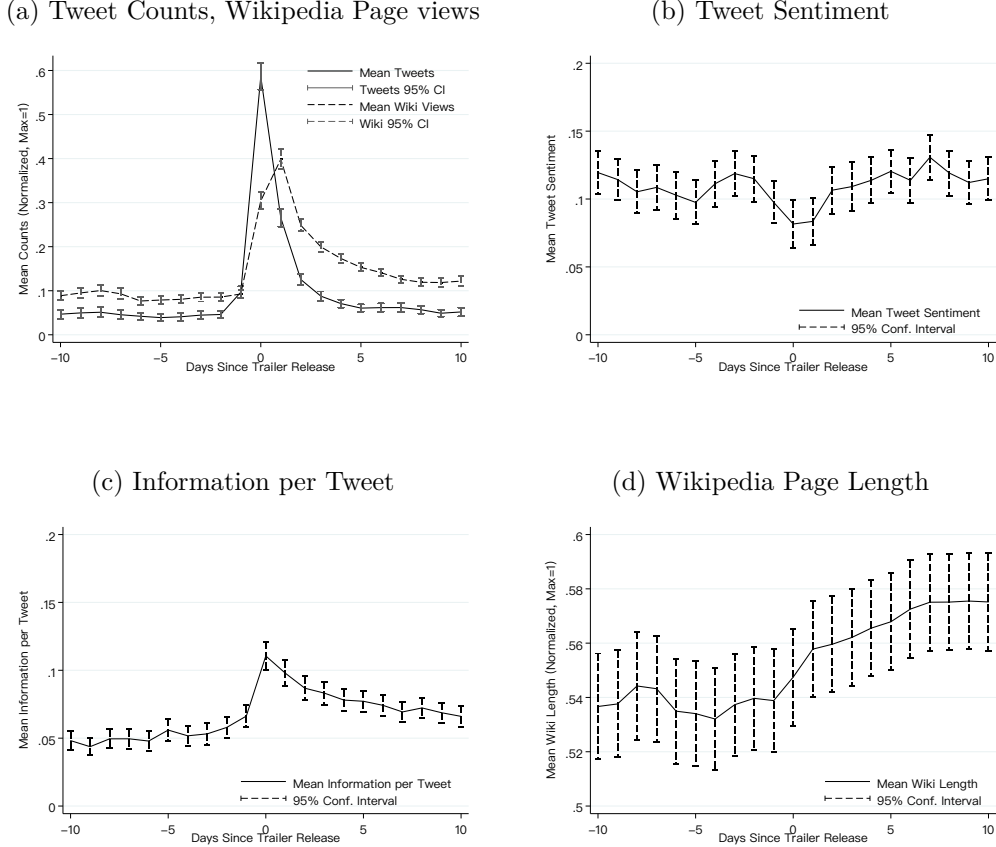
In contrast, Figure 4b shows mean tweet sentiment remains relatively stable throughout the 20-day window, consistent with the insignificant difference observed in the summary statistics. Because sentiment scores range from -1 to 1 , normalization is not necessary. The average sentiment hovers around 0.1 , indicating a slightly positive tone overall.

Figure 4c tracks the average proportion information content per tweet. The informational content of tweets shows a moderate increase following trailer release, suggesting that social media discussions become more content-rich.

⁸We also show in Section 4.2 that there may be no trailers even for movies with a theatrical release, and for these movies there is no trailer price bump.

Figure 4d shows the average Wikipedia page length, normalized by each movie’s maximum observed page length. On average, Wikipedia pages grow slowly on non-trailer days but increase noticeably following a trailer release, reflecting a steady accumulation of new content in response to the promotional material.

Figure 4: Changes in Key Variables by Days since Trailer Release



4.2 Sequels vs. non-sequels

When assessing the impact of movie trailers on market expectations, we distinguish between sequels (or prequels) and non-sequels. Sequels and prequels benefit from existing audience familiarity, meaning potential viewers already hold expectations shaped by earlier films in the franchise. As a result, marketing for these movies tends to reinforce or slightly adjust consumer sentiment rather than introduce entirely new information. In contrast, non-sequels are relatively unknown to audiences, making advertising—especially trailers—more critical in shaping demand expectations. To account for these differences,

we analyze the effects of trailers separately for sequels and non-sequels, providing a clearer view of how new versus established intellectual properties influence market responses.

Among the 580 movies in our sample, 463 are non-sequels. Of all the movies, 111 non-sequels and 9 sequels did not have a trailer released during our observation window prior to their theatrical debut. In total, approximately 20% of movies in our dataset had no trailer release within the sample period.⁹

Table 2: Trailer Timing and HSX Changes by Sequel Status

Days Before Release	Number		Log Change in HSX price	
	Non-Sequel	Sequel	Non-Sequel	Sequel
≤ 30 Days	49	19	0.060*** (0.015)	0.037*** (0.009)
31-60 Days	113	53	0.125*** (0.022)	0.060* (0.031)
61-90 Days	138	38	0.154*** (0.018)	0.037*** (0.006)
> 90 Days	132	35	0.191*** (0.021)	0.060*** (0.013)

Log Change in HSX price on Trailer Days (upto 120 days before theatrical release).

Standard errors in parentheses.

Significantly different from zero, * $p < .1$, ** $p < .05$, *** $p < .01$

Table 2 summarizes how HSX prices respond to trailer releases, distinguishing between sequels and non-sequels and grouping by trailer release timing. We observe 577 trailer release events in total (some movies have multiple trailers). Among movies that released trailers, non-sequels averaged approximately 1.23 trailers per title, compared to 1.34 for sequels. Only about 12% of trailers were released within the final 30 days before theatrical release, indicating that earlier promotional efforts are the norm in industry practice.

The right panel of Table 2 shows that non-sequels tend to experience larger increases in HSX prices following a trailer release compared to sequels. This supports the idea that trailers are more impactful for unfamiliar properties, where they serve to introduce new concepts and generate awareness and excitement. This distinction reinforces the importance of analyzing the two groups separately to accurately assess the effects of

⁹Some of these movies have trailers released prior to our observation window of 120 days before the theatrical date (these cases are analyzed in Appendix A).

advertising on market expectations. Notably, the timing of the trailer release does not significantly affect the magnitude of the price change. On average, trailers increases HSX prices for non-sequels by approximately 15%, while sequels see a more modest increase of around 5%.

5 Results

In this section, we present the main results of our analysis on the impact of movie trailers on HSX prices. Our primary objective is to examine how trailer releases influence demand expectations, measured by the percentage change in HSX prices (logarithmic difference). Specifically, we focus on four distinct effects: the trailer fixed effect, changes in awareness of the movie, changes in sentiment towards the movie, and changes in available information.

5.1 Estimation Results

The main regression specification is,

$$\begin{aligned} \Delta(HSX)_{it} = & \alpha + \beta_0 \text{Trailer}_{it} + \beta_1 \Delta(\text{Aware})_{it} \times \text{Trailer}_{it} + \beta_2 \Delta(\text{Sentiment})_{it} \times \text{Trailer}_{it} \\ & + \beta_3 \Delta(\text{Information})_{it} \times \text{Trailer}_{it} + \gamma_t + \delta_d + \epsilon_{it}. \end{aligned} \quad (1)$$

where i is a movie and t is the number of days until theatrical release. In equation (1) the constant α captures the average daily (log) change in HSX price, while the trailer fixed effect β_0 reflects the change in HSX price on the day of the trailer release. We further examine the interpretation of this effect in Subsection 5.3. The coefficients β_1 , β_2 , and β_3 represent the effects of changes in awareness, consumer sentiment, and informational content following the trailer release, respectively. We also include a time trend γ_t to account for systematic changes in HSX price as the theatrical release date approaches. In addition, we control for day-of-week fixed effects, recognizing that movie-related activity and market responses can potentially cluster around weekends.

The specification only includes interactions, but not direct terms, for awareness, sentiment, and information because our goal is to estimate the effect of their changes specifi-

cally on trailer release days. These variables may fluctuate for many reasons on non-trailer days, but we are interested in whether their changes, when coinciding with a trailer release, influence market expectations. By focusing on interactions, we isolate the trailer-driven component of each variable’s effect on HSX price changes.¹⁰

We do not include movie fixed effects in the regression specification because the dependent variable is the differences in daily HSX prices. By modeling changes rather than levels, we effectively control for time-invariant characteristics of each movie, such as baseline popularity and inherent demand potential. This differencing approach removes the need for movie fixed effects, as any unobserved movie-specific factors that remain constant over time are implicitly accounted for.

We estimate three variations of the model presented in equation (1). The first is a baseline specification that solely measures the impact of trailers and does not include any controls for awareness, sentiment, and information content. This model serves as a reference point to compare to fuller specifications in which the three channels are included. We then extend the model by introducing two alternative sets of controls to capture changes in awareness, sentiment, and information content around trailer releases. In the first specification, awareness is measured by the log change in tweet volume, and information content is captured by the change in the proportion of tweets mentioning key figures associated with the movie, such as actors, directors, and writers. In the second specification, awareness is proxied by the log change in Wikipedia page views, while information content is measured by the log change in Wikipedia page length. In both versions, sentiment is proxied by changes in tweet sentiment scores, constructed using the textual analysis method described earlier.

Table 3 presents the regression results. Columns 1-3 report coefficient estimates from the three model specifications. Columns 4-5 show the corresponding economic effects. We calculate the economic effects of awareness, sentiment, and information by estimating how a one standard deviation increase in each corresponding measure on the trailer release day would affect HSX prices.

The baseline model in column 1 indicates that, on average, HSX prices increase by

¹⁰There is also an empirical reason for omitting these terms directly. Recall from Section 4.1 that there is little change in these variables on non-trailer days.

Table 3: Impact of Awareness, Sentiment, and Information on Log Changes in HSX Price

	Estimates			Economic Effects	
	Baseline	Tweets	Wiki	Tweets	Wiki
Trailer Fixed Effect	0.122***	0.077***	0.060***		
	(0.002)	(0.003)	(0.003)		
<i>Awareness</i>					
$\Delta \ln(\text{Tweets}) \times \text{Trailer}$		0.013*** (0.001)		0.028	
$\Delta \ln(\text{Wiki. Page Views}) \times \text{Trailer}$			0.046*** (0.002)		0.056
<i>Sentiment</i>					
$\Delta \text{ Tweet Sentiment} \times \text{Trailer}$		0.090*** (0.008)	0.102*** (0.008)	0.022	0.024
<i>Information</i>					
$\Delta \text{ Info. per Tweet} \times \text{Trailer}$		0.233*** (0.016)		0.030	
$\Delta \ln(\text{Wiki. Length}) \times \text{Trailer}$			-0.030*** (0.003)		-0.025
Observations	67,996	67,996	67,996		
Day to Release	Yes	Yes	Yes		
Day of Week FE	Yes	Yes	Yes		
Adj. R^2	0.056	0.065	0.065		

Standard errors in parentheses. Significance level – * $p < .1$, ** $p < .05$, *** $p < .01$

approximately 12% on the day of a trailer release. This finding is consistent with the evidence presented in Figure 3. Column 2 presents the specification that uses tweet volume as a proxy for awareness and mentions of key movie personnel per tweet as a proxy for information. Column 3 reports the alternative specification, where Wikipedia page views proxy awareness and Wikipedia page length captures information content. In both models, the fixed effect of trailer releases on HSX prices remains significantly positive, approximately 8% and 6%, respectively. We return to interpreting this effect at the end of the sub-section.

Both awareness proxies significantly increase HSX prices on the day of a trailer release within their respective model specifications. For the “Awareness Effect,” an one standard deviation increase in tweet volume following a trailer release is associated with a 2.8% rise in HSX price, while an one standard deviation increase increase in Wikipedia page

views corresponds to a 5.6% gain.

Changes in tweet sentiment scores have a significant and positive effect on HSX prices across both model specifications. For the “Sentiment Effect,” an one standard deviation increase in tweet sentiment following a trailer release is associated with an approximate 2% increase in HSX price in both models. Recall, however, that trailer-driven sentiment change is about equally likely to be negative, so for many movies this effect decreases expected box office.

For the “Information Effect,” we find that the information-per-tweet proxy has a positive and significant impact on HSX prices on the day of a trailer release, while changes in Wikipedia page length have a negative and significant effect. In terms of economic magnitude, a one standard deviation increase in tweet information content is associated with a 3% rise in HSX price, whereas an increase in Wikipedia page length corresponds to a 2.5% decline.

This contrasting result highlights a key nuance in how different forms of information engagement influence market expectations. The positive effect of tweet information content suggests that trailers often generate excitement on social media, where users repeatedly mention cast and directors in a way that amplifies buzz, even if the content itself can be somewhat redundant. In contrast, Wikipedia activity reflects a more deliberate, information-seeking behavior. A surge in Wikipedia page length may not always signal market optimism; rather, it might reflect the disclosure of additional plot details, production challenges, or genre-specific information that tempers expectations. Thus, more information is not necessarily better, at least from the perspective of market expectations.¹¹

Importantly, the strongest driver remains the trailer fixed effect, even after controlling for awareness, consumer sentiment, and informational content. In essence, trailers help reduce uncertainty about a movie’s commercial trajectory. They provide a credible and visible milestone that reassures both consumers and market participants that the film

¹¹One concern might be that our measure of change in Wikipedia page length may disadvantage movies that already have lengthy pages prior to trailer release, since the scope for further expansion is more limited. To address this, we control for pre-trailer Wikipedia page length in our regression, and the results remain robust. We also assess potential multicollinearity between Wikipedia page views and Wikipedia length changes using variance inflation factors (VIF). The VIF values are below 5, indicating that multicollinearity is not a significant concern in our specifications.

will reach theaters rather than being delayed, canceled, or quietly diverted to streaming or direct-to-video platforms.¹² In this way, the trailer functions not just as marketing content but as a public commitment device, anchoring market expectations and validating the movie’s status within the release pipeline. We will examine how trailer releases help reduce uncertainty in Section 5.3.

5.2 Sequels vs. Non-Sequels

Our analysis shows that trailer releases significantly shape market expectations of movies. However, it is important to distinguish between sequels/prequels and non-sequels, as trailers appear to influence these two categories differently. As discussed in Section 4.2, trailers on average increases HSX prices for non-sequels by approximately 15%, while sequels see a more modest increase of around 5%

Sequels typically benefit from pre-existing familiarity and greater awareness of the franchise. Potential viewers often arrive with preconceived notions based on prior films, and their sentiment is shaped by prior experiences with the series. As a result, trailers for sequels may play a more limited role in shifting awareness, sentiment, or providing new information.changes in tweet volume, Wikipedia page views, or sentiment.

In contrast, non-sequels, original films without an established audience, rely more heavily on trailers to generate interest, shape perceptions, and communicate essential information about the film’s genre, tone, and key talent. For these films, trailers play a critical role in generating first impressions and shaping audience awareness. As a result, we observe stronger trailer-induced changes in public attention and information-seeking behavior for non-sequels.

Tables 4 and 5 report results from estimating equation (1) separately for sequels and non-sequels (Table 4 shows the point estimates and Table 5 the economic effect). By analyzing sequels and non-sequels separately, we can better understand how changes in HSX prices, sentiment, and Wikipedia activity vary with the level of prior consumer knowledge and engagement. This distinction allows for a more precise understanding

¹²For example, many films that enter late-stage production or even complete filming are ultimately shelved or released directly to streaming without a theatrical run (see Section 4.1). The trailer, especially when widely distributed and professionally produced, signals that a studio is investing in theatrical promotion and release infrastructure.

Table 4: Impact of Awareness, Sentiment, and Information on Log Changes in HSX Price by Sequel Status

	Non-Sequel		Sequel	
	Tweets	Wiki	Tweets	Wiki
Trailer Fixed Effect	0.096*** (0.004)	0.069*** (0.004)	0.040*** (0.003)	0.058*** (0.003)
<i>Awareness</i>				
$\Delta \ln(\text{Tweets}) \times \text{Trailer}$	0.014*** (0.001)		0.005*** (0.001)	
$\Delta \ln(\text{Wiki. Page Views}) \times \text{Trailer}$		0.055*** (0.002)		-0.006*** (0.002)
<i>Sentiment</i>				
$\Delta \text{ Tweet Sentiment} \times \text{Trailer}$	0.084*** (0.010)	0.097*** (0.009)	0.042*** (0.011)	0.065*** (0.011)
<i>Information</i>				
$\Delta \text{ Info. per Tweet} \times \text{Trailer}$	0.187*** (0.018)		0.173*** (0.034)	
$\Delta \ln(\text{Wiki. Length}) \times \text{Trailer}$		-0.042*** (0.004)		0.025*** (0.003)
Observations	54,054	54,054	13,942	13,942
Day to Release	Yes	Yes	Yes	Yes
Day of Week FE	Yes	Yes	Yes	Yes
Adj. R^2	0.070	0.073	0.052	0.054

Standard errors in parentheses. Significance level: * $p < .1$, ** $p < .05$, *** $p < .01$

of how trailer releases influence market expectations and investor confidence in a film’s commercial prospects.

Trailer fixed effects are statistically significant in all models, but the magnitude is notably larger for non-sequels: trailer releases raise HSX prices by 7–10% for non-sequels, compared to 4–6% for sequels. We have argued that this term largely reflects a commitment signal about a theatrical release, which is more important for non-franchise movies. Section 5.3 provides more evidence on this point.

The “Awareness Effect” varies significantly by sequel status as expected. For non-sequels, an one standard deviation increase in tweet volume or Wikipedia page views after trailer release corresponds to substantial HSX price increases (about 3% and 7%, respectively). In contrast, the same changes yield negligible effects for sequels, likely because awareness was already high for sequels before the trailer release.

Table 5: Estimated Economic Effects of Awareness, Sentiment, and Information by Sequel Status

Models	Non-Sequel		Sequel	
	Tweets	Wiki	Tweets	Wiki
<i>Awareness</i>				
$\Delta \ln(\text{Tweets}) \times \text{Trailer}$	0.030		0.011	
$\Delta \ln(\text{Wiki. Page Views}) \times \text{Trailer}$		0.066		-0.007
<i>Sentiment</i>				
$\Delta \text{Tweet Sentiment} \times \text{Trailer}$	0.020	0.023	0.010	0.016
<i>Information</i>				
$\Delta \text{Info. per Tweet} \times \text{Trailer}$	0.024		0.022	
$\Delta \ln(\text{Wiki. Length}) \times \text{Trailer}$		-0.035		0.020

The “Sentiment Effect” is more consistent across both groups. An one standard deviation increase in tweet sentiment following trailer release corresponds to a roughly 1–2% increase in HSX price, regardless of whether the movie is a sequel or not. This is consistent with our earlier claims that trailers reveal movie quality, both positive and negative, which should have a comparable effect on consumer demand for both sequels and non-sequels.

The information-per-tweet measure of the “Information Effect” has a near identical effect for sequel and non-sequel films. An one standard deviation increase in this measure raises HSX prices by approximately 2% in both categories, suggesting that more frequent mentions of key figures, such as actors or directors, within tweets generate broadly positive market signals, regardless of a film’s sequel status.

In contrast, the effect of Wikipedia page length changes diverges significantly. For non-sequels, an one standard deviation increase in Wikipedia length corresponds to a 3.5% decline in HSX price, while for sequels, it is associated with a 2% increase.

As discussed earlier, longer Wikipedia pages can reveal both positive and negative aspects of a movie, making it a priori ambiguous whether greater information disclosure improves or dampens market expectations. For non-sequels, additional information may temper early enthusiasm or introduce uncertainty about the film’s appeal, especially when audiences are unfamiliar with the concept or cast. In contrast, for sequels – where audiences already have a baseline level of familiarity – expanded Wikipedia entries in these cases may serve to confirm continuity with the original film, thereby increasing rather

than weakening consumer demand.

Taken together, these results highlight how the impact of trailers varies depending on a movie’s baseline familiarity. While awareness and information effects can differ sharply between sequels and non-sequels, the trailer fixed effect remains large and significant – particularly for non-sequels. This persistent effect, even after accounting for measurable changes in attention and sentiment, points to a broader role that trailers may play in shaping expectations. In what follows, we explore whether this effect reflects a reduction in market uncertainty, especially for films without established franchise support.

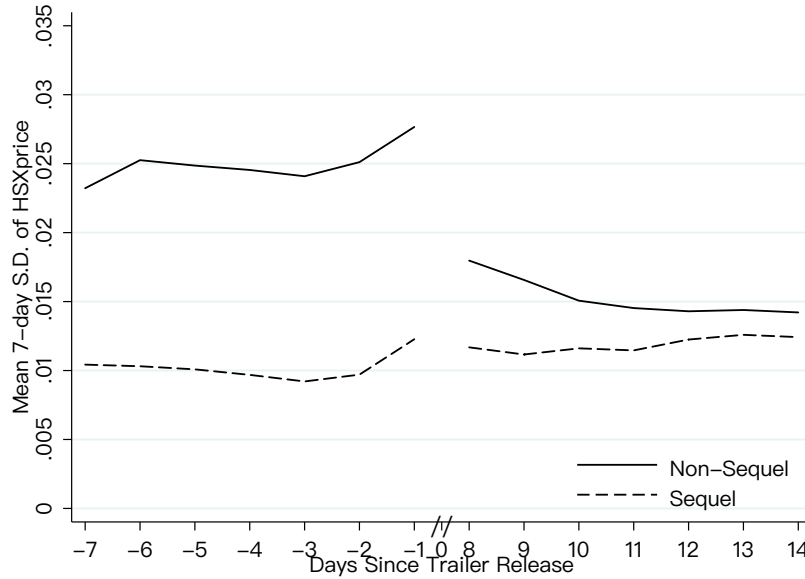
5.3 Uncertainty Reduction

In the main regression (Table 3), we find a significant and economically meaningful trailer fixed effect. In the subgroup analyses for sequels and non-sequels (Tables 4 and 5), this effect is notably larger for non-sequels. We propose that this stronger fixed effect for non-sequels reflects a greater reduction in uncertainty following a trailer release. Specifically, we argue that trailers serve as key market signals, reassuring investors and consumers about a film’s theatrical prospects and thereby increasing demand expectations. In this section, we empirically test this explanation by examining whether uncertainty, as measured by HSX price volatility, declines after a trailer release, and whether this effect differs between sequels and non-sequels.

Measuring movie demand uncertainty is essential to understanding how market expectations evolve before a film’s release and how external signals, such as trailers, influence investor and consumer perceptions. Prior to a trailer release, there is often significant uncertainty surrounding a movie’s commercial viability, including whether it will secure a theatrical release or be relegated to streaming or direct-to-video. A significant share of movie titles listed on HSX never reach theaters, causing their HSX prices to drop to zero as the market adjusts its expectations.

Although our sample includes only wide-released films, uncertainty about their theatrical prospects often remains prior to the trailer debut. Investors may question whether a film will secure a full theatrical release or shift to alternative distribution channels. The trailer acts as a crucial market signal, confirming the film’s readiness for theatrical release

Figure 5: HSX Price Volatility



and reducing uncertainty. This likely explains the strong trailer fixed effect observed in our regression results, as HSX prices tend to rise significantly following trailer releases.

To proxy for market uncertainty surrounding a movie’s potential performance, we use HSX volatility.¹³ Higher volatility reflects greater fluctuations in market expectations of consumer demand, indicating increased investor uncertainty. Conversely, lower volatility implies more stable market expectations and greater confidence in a movie’s box office trajectory. We define HSX price volatility as the standard deviation of a movie’s HSX price over a rolling seven-day window leading up to and including the current day. To account for differences in price levels across movies, we normalize this standard deviation by the mean HSX price over the same seven-day period. This normalization enables consistent comparison of volatility across films with varying baseline prices.

Figure 5 displays the mean seven-day rolling normalized standard deviation of HSX prices, plotted relative to the trailer release date (day 0).¹⁴ The two lines represent trends for sequels (and prequels) and non-sequels, allowing for a direct comparison of how HSX

¹³An alternative forward-looking measure would be implied volatility, typically derived from option prices using the Black-Scholes formula. However, HSX puts and calls are introduced only shortly before a movie’s release, making them unsuitable for measuring expected volatility in the earlier pre-release window that is our focus.

¹⁴The figure omits seven days immediately following a trailer release, as the associated price spike can temporarily inflate HSX volatility without reflecting underlying uncertainty.

price volatility evolves around trailer releases.

Prior to trailer release, non-sequels exhibit higher volatility, indicating greater uncertainty about their commercial prospects. After the trailer release, these movies have a noticeable decline in volatility, suggesting that trailers serves as important signals, reducing uncertainty by providing key information about the film’s production quality, genre, and likelihood of a wide release.

In contrast, the release of trailers have little impact on the volatility of sequels, indicating that consumers and investors already have greater confidence in their theatrical prospects. This is likely because sequels’ association with established franchises and built-in fan bases, which reduce ambiguity around release plans.

Notably, after the trailer release, volatility of non-sequels drops to the same level as for sequels, indicating that the trailer effectively closes uncertainty gap between the two. This convergence suggests that once a trailer is released, investors and consumers adjust their expectations accordingly, viewing non-sequels with a similar level of confidence as franchise films. This pattern supports our hypothesis that trailers play a crucial role in resolving market uncertainty, helping to explain the strong trailer fixed effect observed in our regression results.

We further examine how trailer releases influence HSX price volatility while controlling for movie-level confounding factors using the following movie-fixed-effect regression:

$$SD(HSX)_{it} = \mu_i + \phi_1 \text{PostTrailer}_t + \phi_2 \text{Sequel}_i + \phi_3 (\text{PostTrailer}_t \times \text{Sequel}_i) + \epsilon_{it} \quad (2)$$

where $SD(HSX)_{it}$ denotes the seven-day rolling standard deviation of HSX prices for movie i at day t , capturing fluctuations in market expectations. The key explanatory variable, PostTrailer_t , is a binary indicator equal to one for periods beginning after the trailer release—excluding the first seven days—and continuing through two weeks post trailer release.. The variable Sequel_i is an indicator that equals one if the movie is a sequel, while the interaction term $\text{PostTrailer}_t \times \text{Sequel}_i$ captures any differential post-trailer volatility response for sequels relative to non-sequels. The regression excludes observations from the seven-day window immediately following the trailer release to avoid short-term price spikes. For movies without a trailer, PostTrailer_t remains zero for all observations.

Table 6: HSX Price Volatility Regression, equation (2)

	Volatility of HSX Price
Post Trailer	−0.0032*** (0.0006)
Post Trailer × Sequel	0.0046*** (0.0013)
Post trailer+ Post trailer× Sequel	0.0014 (0.0011)
Movie fixed effect	Yes
No. of Movies	580

Standard errors in parentheses. Significance Level: * $p < .1$, ** $p < .05$, *** $p < .01$

The term μ_i represents movie fixed effects, absorbing time-invariant differences across films.

Table 6 confirms that trailers releases significantly reduce HSX price volatility for non-sequels but have relatively little effect on sequels. The *PostTrailer* coefficient is negative and highly significant, indicating a substantial decline in volatility of non-sequels after a trailer release. In contrast, the positive and significant interaction term suggests that this effect is much weaker for sequels. The combined effect ($PostTrailer + PostTrailer \times Sequel$) is small and not significant, implying that sequels’ volatility remains largely unchanged after the trailer.

These findings support our hypothesis that trailers resolve uncertainty primarily for non-sequels, which initially face greater market unpredictability. For sequels, where expectations are more firmly established, trailers have a limited impact on volatility. The inclusion of movie fixed effects ensures that these results are not driven by inherent differences across films.

6 Conclusion

This paper provides a detailed empirical decomposition of how advertising influences consumer behavior and market expectations, focusing specifically on the movie industry through the unique lens of movie trailer releases. By leveraging a rich dataset of nearly 600 wide-release U.S. films from 2017 to 2021, the study identifies three distinct

pathways—raising awareness, shaping sentiment, and conveying information—and empirically isolates their individual effects. Key findings demonstrate that trailer releases—a discrete advertisement—significantly increase pre-theatrical market expectations, especially for non-sequel films, as reflected by substantial rises in Hollywood Stock Exchange (HSX) prices. Moreover, the analysis reveals that while increased consumer engagement through Twitter and Wikipedia correlates positively with market expectations, detailed informational content provided through trailers can occasionally dampen enthusiasm.

From a managerial perspective, the results offer valuable insights for marketing professionals and film industry stakeholders. Managers should strategically leverage trailers as tools not only to boost consumer awareness and positive sentiment but also to carefully manage the level of informational detail provided, as excessive explicit information may inadvertently temper audience enthusiasm. Additionally, trailers play a crucial role in reducing market uncertainty, particularly for novel films, highlighting their importance in strategic promotional planning. Thus, firms are encouraged to tailor their advertising content to maximize engagement and optimize consumer response in an increasingly complex and digitally driven market environment.

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Appendix A Robustness to Trailer Release Timing

Our analysis focuses on the four month window prior to a movie’s theater date, primarily due to Twitter availability constraints (we have far longer data for our other variables—including HSX price, Wikipedia page views, and Wikipedia page length—and also have the full list of trailer release dates).¹⁵ A potential concern is that some films release trailers earlier, and if such trailers affect market expectations differently than those released closer to the theater date, this limitation could introduce bias into our estimates.

We provide three pieces of evidence which are inconsistent with this claim. Movies with earlier trailers are observationally similar. When we expand the time window to also include trailers released more than 120 days before the theater date, the main estimates are largely unchanged (this is for a sparser specification which omits Twitter variables). And finally to see how sensitive our results are to the observation window we can go in reverse: the full specification is largely unchanged when we reduce our data from their current four month pre-theater period.

Table A1 compares the descriptive characteristics of movies across four trailer release categories. The first category includes movies with no trailer prior to release. The second group released a trailer only more than 120 days before release, while the third group released a trailer only within 120 days. The final group had trailers released in both time periods. Our analysis omits movies from the second group and a part of the fourth.

The main takeaway is that these two omitted groups are largely similar to the others across a variety of observed characteristics.¹⁶ This suggests their omission is not likely to drive our results. In particular, the distribution of genres and sequel status is relatively similar across movies with trailers. The Bechdel rating, which measures gender representation, also shows little variation across groups.

The second set of robustness results extend the observation window to include all trailers, including those released prior to 120 days, in a version of the main specification equation (1). We replicate the models in Table 3 that incorporate all available HSX price

¹⁵Twitter data is limited to tweets posted within this period because changes to the platform’s API policy have restricted academic access to historical tweet data.

¹⁶Movies without trailers differ notably from those with trailers. This is not an issue since they are included in the sample.

Table A1: Movie Characteristics by Trailer Release Timing Category (Including trailers more than 120 days before theatrical date)

	No Trailer	Trailer > 120 Days Days Only	Trailer $\leq$ 120 Days Only	Trailer in both Periods	Overall
Sequel	0.080 (0.271)	0.116 (0.320)	0.221 (0.415)	0.380 (0.485)	0.282 (0.450)
Comedy	0.256 (0.437)	0.165 (0.372)	0.159 (0.365)	0.134 (0.341)	0.152 (0.359)
Action	0.220 (0.414)	0.256 (0.436)	0.251 (0.434)	0.357 (0.479)	0.303 (0.460)
Drama	0.426 (0.494)	0.447 (0.497)	0.357 (0.479)	0.288 (0.453)	0.336 (0.472)
Horror	0.078 (0.268)	0.118 (0.323)	0.120 (0.326)	0.0752 (0.264)	0.096 (0.294)
Thriller	0.109 (0.312)	0.024 (0.152)	0.077 (0.267)	0.019 (0.137)	0.044 (0.205)
Animated	0.001 (0.029)	0.047 (0.211)	0.025 (0.156)	0.091 (0.287)	0.059 (0.236)
Bechdel Rating	2.077 (1.211)	2.514 (0.848)	2.268 (0.954)	2.538 (0.803)	2.443 (0.882)
Num. of Movies	54	66	226	234	580

mean coefficients; sd in parentheses

data and Wikipedia data (but omit Twitter variables). Two specifications are estimated: (1) the baseline model from Table 3, and (2) a model that includes only Wikipedia-based proxies for awareness and information content. For both of these we have data extending beyond the 120-day cutoff for all variables.

Table A2 contains the estimates. Column 1 replicates the baseline model from Table 3, using data limited to the 120-day window before movie theatrical release. Column 2 extends the sample to include data beyond 120 days, estimating the same baseline model. The results show that the average log change in HSX prices on trailer release days remains nearly identical across both columns, suggesting the expanded time window does not materially alter the baseline estimate.

In column 3, we estimate the model including as regressors Wikipedia page view and page length measures while applying the 120-day cutoff.¹⁷ Column 4 has the same specification but includes data beyond the 120-day window. The parameter estimates are almost identical to those from column 3, further supporting the robustness of our findings.

¹⁷Although this specification excludes the Twitter sentiment measure, the parameter estimates on the included variables are virtually unchanged with those reported in Table 3.

Table A2: Robustness – Trailers Released More than 120 Days Before Theatrical Date (Dependent Variable = Log Changes in HSX Price)

	Baseline		Wiki Measures		Economic Effects	
	≤ 120 Days	All	≤ 120 Days	All	≤ 120 Days	All
Trailer Fixed Effect	0.122*** (0.002)	0.121*** (0.002)	0.060*** (0.003)	0.073*** (0.003)		
<i>Awareness</i>						
$\Delta \ln(\text{Wiki. Page Views}) \times \text{Trailer}$			0.046*** (0.002)	0.031*** (0.001)	0.056	0.046
<i>Information</i>						
$\Delta \ln(\text{Wiki. Length}) \times \text{Trailer}$			-0.031*** (0.003)	-0.023*** (0.002)	-0.026	-0.024
Observations	67,996	221,785	67,996	221,785		
Day to Release	Yes	Yes	Yes	Yes		
Day of Week FE	Yes	Yes	Yes	Yes		
Adj. R^2	0.056	0.026	0.063	0.028		

Standard errors in parentheses. Significance level – * $p < .1$, ** $p < .05$, *** $p < .01$

Using these estimates, columns 5-6 compare the economic effects of changes in awareness and information content on trailer release days. An one standard deviation increase in the corresponding Wikipedia proxy measures leads to very similar changes in HSX prices, regardless of whether the 120-day cutoff is applied. These results reinforce our confidence that the main findings are not biased due to the specific time window used in the analysis.

The final set of results are based on a different strategy to assess the sensitivity to the time window. Instead of increasing the window, we instead shrink it. This allows us to include all variables from our main specification, including the Twitter-based ones, which will give a richer set of parameter estimates to compare. A telltale sign our results are not robust would be if shortening the window markedly changes the estimates.

Table A3 presents estimates for the Twitter-based models from Table 3, using a more restrictive 90-day cutoff window as well as the original 120-day one. Columns 2 and 4 of replicate the corresponding columns in Table 3, using all Twitter-based variables – changes in tweet volume, sentiment scores, and information content – within 120 days before a movie’s release. Columns 1 and 3 re-estimate the same models using only data from within 90 days before release.

The results show that estimates are broadly consistent across the two cutoff windows. The trailer fixed effects are nearly identical, while the awareness, sentiment and infor-

Table A3: Robustness – Shortening Window to 90 Days Before Theatrical Date (Dependent Variable = Log Changes in HSX Price)

	Tweet Measures		Economic Effects	
	≤90 Days	≤120 Days	≤90 Days	≤120 Days
Trailer Fixed Effect	0.061*** (0.005)	0.060*** (0.003)		
<i>Awareness</i>				
$\Delta \ln(\text{Tweets}) \times \text{Trailer}$	0.019*** (0.002)	0.013*** (0.001)	0.036	0.028
<i>Sentiment</i>				
$\Delta \text{ Tweet Sentiment} \times \text{Trailer}$	0.073*** (0.012)	0.102*** (0.008)	0.016	0.024
<i>Information</i>				
$\Delta \text{ Info. per Tweet} \times \text{Trailer}$	0.295*** (0.023)	0.233*** (0.016)	0.036	0.030
Observations	40,746	67,996		
Day to Release	Yes	Yes		
Day of Week FE	Yes	Yes		
Adj. R^2	0.054	0.065		

Standard errors in parentheses. Significance level – * $p < .1$, ** $p < .05$, *** $p < .01$

mation estimates are quite similar. As with the other robustness checks, these patterns suggest that our main findings are not sensitive to the specific cutoff window used.

Appendix B Robustness to COVID Restriction

Our main sample includes movies released in both 2020 and 2021, which overlaps with the COVID-19 pandemic period. A potential concern is that the pandemic introduced substantial external uncertainty that could affect both market expectations and the effectiveness of promotional activities such as trailer releases. To address this, we conduct a robustness check by restricting the sample to movies released before March 1, 2020. Table A4 columns 1-3 replicate the corresponding columns from Table 3. Columns 4-6 present estimates using the restricted pre-COVID sample.

Comparing Columns 1 and 4, the baseline estimates of HSX price changes due to trailer releases remain largely consistent. In the two model specifications (columns 2-3 and 5-6) that include controls for awareness, sentiment, and information, we find that the trailer fixed effects are slightly larger in the pre-COVID period.¹⁸ The parameters on the awareness, sentiment and information variables are quite similar whether or not data for the last years are included. In short the results are robust to the removal of movies released during COVID.

¹⁸As discussed in Section 5.3, trailer releases play a key role in reducing uncertainty about a movie's theatrical prospects. During the pandemic, however, elevated external uncertainty – especially surrounding public lockdown and the likelihood of theatrical release – introduced risks that trailers alone could not offset, potentially weakening their impact on market expectations.

Table A4: Robustness — Omit Movies Released During COVID (Dependent Variable = Log Changes in HSX Price)

	All Periods			Pre-COVID Periods		
	Baseline	Tweets	Wiki	Baseline	Tweets	Wiki
Trailer Fixed Effect	0.122*** (0.002)	0.077*** (0.003)	0.060*** (0.003)	0.125*** (0.002)	0.082*** (0.003)	0.068*** (0.004)
<i>Awareness</i>						
$\Delta \ln(\text{Tweets}) \times \text{Trailer}$		0.013*** (0.001)			0.013*** (0.001)	
$\Delta \ln(\text{Wiki. Page Views}) \times \text{Trailer}$			0.046*** (0.002)			0.044*** (0.002)
<i>Sentiment</i>						
$\Delta \text{ Tweet Sentiment} \times \text{Trailer}$		0.090*** (0.008)	0.102*** (0.008)		0.090*** (0.009)	0.099*** (0.009)
<i>Information</i>						
$\Delta \text{ Info. per Tweet} \times \text{Trailer}$		0.233*** (0.016)			0.226*** (0.018)	
$\Delta \ln(\text{Wiki. Length}) \times \text{Trailer}$			-0.030*** (0.003)			-0.034*** (0.003)
Observations	67,996	67,996	67,996	56,785	56,785	56,785
Day to Release	Yes	Yes	Yes	Yes	Yes	Yes
Day of Week FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.056	0.065	0.065	0.059	0.067	0.065

Standard errors in parentheses. Significance level – * $p < .1$, ** $p < .05$, *** $p < .01$